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Key Points:

- Variations in solar output left an imprint on climate during the pre-industrial Holocene, detectable using a nonlinear causality method
- Influence is observed in both global and high-latitude temperature single-site records, but it is clearer in the latter
- The detected solar imprint could serve as a benchmark for testing whether climate models capture such subtle dynamics

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

J. P. Landers,
lpander@usc.edu

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Author Contributions:

Conceptualization: Jordan P. Landers, Julien Emile-Geay, Deborah Khider, Edouard Bard

Formal analysis: Jordan P. Landers, Edouard Bard

Funding acquisition: Julien Emile-Geay, Deborah Khider

Investigation: Jordan P. Landers, Alexander K. James, Stephan B. Munch, Deborah Khider, Edouard Bard

Methodology: Jordan P. Landers, Alexander K. James, Stephan B. Munch, Deborah Khider

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A Causal Examination of the Solar Influence on Holocene Climate

Jordan P. Landers¹ , Julien Emile-Geay¹ , Alexander K. James^{1,2} , Stephan B. Munch³, Deborah Khider⁴ , and Edouard Bard⁵ 

¹Department of Earth Sciences, University of Southern California, Los Angeles, CA, USA, ²Deep Blue Geophysics, Los Angeles, CA, USA, ³Department of Applied Mathematics, University of California, Santa Cruz, CA, USA, ⁴Information Sciences Institute, University of Southern California, Marina del Rey, CA, USA, ⁵CEREGE, Aix-Marseille University CNRS, IRD, INRAE, Collège de France, Aix-en-Provence, France

Abstract Variations in Total Solar Irradiance (TSI) have long been hypothesized to influence global mean surface temperature (GMST) on multidecadal to millennial timescales, but empirical evidence has remained elusive. Here, we explore the link between solar variability and climate over the pre-industrial Holocene, when anthropogenic influences were minimal, using the causal inference framework of Convergent Cross-Mapping. We analyze pairings of two temperature targets (a GMST reconstruction and the central Greenland GISP2 record) with two TSI reconstructions across a range of timescales. Our results indicate a discernible influence of TSI on temperature in ~80% of parameter configurations: stronger but less consistently significant for GISP2, weaker but consistently significant for the global mean. This suggests that solar forcing was a minor—but detectable—driver of global Holocene climate fluctuations, an empirical insight that may serve as a benchmark for evaluating whether climate models correctly capture such subtle, nonlinear dynamics.

Plain Language Summary Understanding how variation in a natural forcing such as solar brightness affects climate is important for interpreting today's changing climate and predicting its future trajectory. Here we investigate whether solar brightness influenced climate over the past several thousand years using Convergent Cross-Mapping, a method for testing causal relationships. We detect a subtle, but consistent solar influence at the global scale, and a clearer influence in a high-latitude record. This result could provide a benchmark for evaluating whether climate models correctly capture the full response to solar forcing, one of Earth's most fundamental climate drivers.

1. Introduction

Natural climate variability, whether endogenous or exogenous, can amplify or offset anthropogenic warming. Thus, improving climate predictions requires understanding all factors driving this variability. Because the Sun energizes nearly all climate phenomena, variations in solar brightness (total solar irradiance, or TSI) have long been hypothesized as a key climate driver (e.g., Langley, 1876). Space-based radiometric observations over the last several decades have documented ~0.1% variations in TSI, corresponding to the 11-year Schwabe cycle (Schwabe, 1844), which has been linked to temperature fluctuations on the order of .05–.1°C (Amdur et al., 2021; Meehl et al., 2009; Misios et al., 2016; White et al., 1997). The observed temperature response to the Schwabe cycle, consistent with Earth system model simulations (Amdur et al., 2021; Misios et al., 2016), establishes that solar forcing accounts for only a minor contribution to industrial-era warming (Amdur & Huybers, 2025; Intergovernmental Panel on Climate Change (IPCC), 2023; Yeo et al., 2020).

Various records of solar activity prior to the satellite era (e.g., sunspot numbers, cosmogenic nuclides) suggest fluctuations at multi-decadal to millennial scales (Figure 1b): the ~88 years Gleissberg cycle (Gleissberg, 1939; Peristykh & Damon, 2003), the ~205 years DeVries-Suess cycle (de Vries, 1958; Suess, 1965), and the ~2400 years Hallstatt cycle (Damon & Sonett, 1991; Usoskin et al., 2016). However, it is less clear whether these fluctuations produce detectable climate variations (e.g., Board, 1982; Denton & Karlén, 1973; Eddy et al., 1982; Haigh, 2002; Lean & Rind, 1999; Rind, 2002), and if they do, whether they operate via the same dynamics as the Schwabe cycle.

Beyond direct radiative heating, the favored mechanism linking solar fluctuations to climate involves ultraviolet radiation-induced changes in stratospheric ozone that affect the polar vortex and jet streams, with cascading

Project administration: Julien Emile-Geay
Software: Jordan P. Landers, Deborah Khider
Supervision: Julien Emile-Geay, Deborah Khider
Visualization: Jordan P. Landers, Alexander K. James
Writing – original draft: Jordan P. Landers
Writing – review & editing: Jordan P. Landers, Julien Emile-Geay, Alexander K. James, Stephan B. Munch, Deborah Khider, Edouard Bard

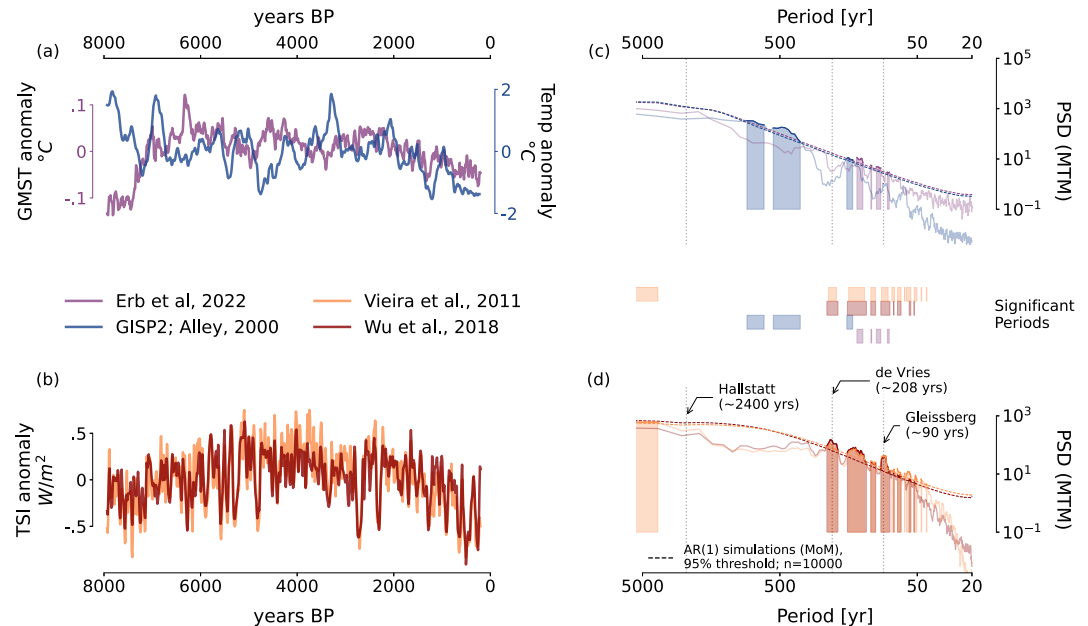


Figure 1. (Left) Time series (a) temperature anomaly: ALLEY00 (navy) scaled to the right y-axis, ERB22 (purple) scaled to the left y-axis, both shown in years BP (b) total solar irradiance (TSI) anomaly: VIEIRA11 (orange) and WU18 (red) both scaled to the same y-axis; (Right) spectral density across a range of periods, with significant periods relative to AR(1) surrogates (Khider et al., 2022) highlighted with shaded columns, (c) temperature targets (ERB22 and GISP2), (d) TSI reconstructions (VIEIRA11, WU18) with annotations indicating de Vries and Gleissberg cycles. Significant periodicities highlighted on the common axis between (c) and (d) indicate considerable overlap between the two TSI reconstructions, but that the temperature targets exhibit distinct spectral patterns and unique overlap with TSI bands.

impacts on tropospheric circulation (Thiéblemont et al., 2015). Although surface-level effects remain speculative (Chiodo et al., 2019), some studies suggest that solar variability may influence ocean heat uptake, circulation patterns, and the ice-albedo feedback (Gray et al., 2010), as well as internal climate modes, including the El Niño–Southern Oscillation (Emile-Geay et al., 2007), which in turn has been linked to monsoon variability (Chen et al., 2024).

Whatever the mechanism, a longstanding question in climate studies boils down to: “Does the Sun drive detectable variations in surface temperature over multi-decadal to millennial timescales?” (Bard & Frank, 2006; Rind, 2002). This study addresses this question by applying a causal discovery framework (Convergent Cross-Mapping, or CCM (Sugihara et al., 2012)) to different estimates of TSI and temperature spanning the last ~8,000 years of the pre-industrial Holocene (last ~11,600 years of Earth’s history). First, we offer context around establishing causality with paleoclimate time series and provide a general overview of the CCM framework. Second, we outline the design of the study, introducing the data sets and our approach to analyzing them using CCM. In the third section, we present our results and discuss them in the fourth.

2. Causality in Paleoclimate Timeseries

Paleoclimate studies from a range of locations (e.g., the North Atlantic: (Bond et al., 2001; Jiang et al., 2015), Oman: (Neff et al., 2001), the Arabian Sea: (Jung et al., 2002), the Northeastern Pacific (Galloway et al., 2013)) have attributed past climate variations to low-frequency changes in TSI, primarily on the basis of spectral similarities between proxy records and TSI reconstructions. A separate set of studies observe variable lead-lag patterns between TSI and their records, hypothesizing that although TSI may have influence, the signal of this relationship is contorted as it is expressed through layers of local and regional-scale processes (Laurenz et al., 2019; Mauquoy et al., 2008; Turney et al., 2005). Still others (e.g., Bengtsson, 2006; Pittcock, 1978; Turner et al., 2016) argue that shared spectral characteristics alone do not establish causality because internal climate variability can reasonably produce periodicities that resemble TSI without any direct solar influence. We therefore require a technique better suited to distinguishing causality from coincidence in the presence of nonlinear and indirect influences.

Nonlinear systems, such as Earth's climate, evolve under the influence of internal states, feedback mechanisms, and external drivers. Because the system's response depends on its current state, these influences jointly shape its trajectory, and the trajectory of each observable variable encodes their collective imprint. A key result from Takens's theorem (Takens, 1981) formalizes this idea: under certain conditions, the full dynamics of a system can be reconstructed from the time series of a single observed variable. Thus, if two variables are part of the same dynamical system, reconstructing the dynamics of either recovers information about both; each time series contains a version of the other's story. Based on Takens's insight, if X influences Y , traces of X 's dynamics should be retrievable from Y 's time series, including in systems that involve independent deterministic forcings (Stark, 1999; Stark et al., 2003). This holds, provided the causal influence of X is sufficiently persistent. Transient forcing (e.g., volcanic eruptions) or purely stochastic drivers do not satisfy this criterion and are therefore outside the scope of this framework established by Takens and extended by Stark.

In the absence of well-known governing equations, any link between forcing and response must be inferred empirically from time series data (Runge et al., 2019). Empirical Dynamical Modeling (EDM) leverages this principle to predict system behavior directly from observed data. Instead of requiring explicit equations or predefined functional forms, EDM uses pattern matching to identify historical states similar to the context of interest and extrapolates their outcomes to predict future states (Sugihara & May, 1990). This approach makes nonlinear variables predictable from their own past states, but predictability alone does not establish causality.

In these systems, causally connected variables may not show consistent correlation patterns; the same nonlinear system can exhibit periods of positive, negative, and no correlation between coupled variables (Sugihara et al., 2012). Convergent Cross-Mapping (CCM, (Sugihara et al., 2012)) extends EDM by offering a framework for assessing causality between variables in nonlinear systems using just time series. Specifically, CCM tests this by quantifying whether the state history of one variable Y (or its proxy) contains enough information to accurately reconstruct the historical state of another variable X (or its proxy). The precision of this reconstruction, "CCM skill," is measured by the correlation (ρ) between actual and estimated values (Sugihara et al., 2012).

CCM avoids the pitfalls of simply correlating X and Y by requiring both significant predictive skill and convergence—the systematic improvement of predictions with increasing sample size. Systematic improvement indicates that additional examples (viz., an increase in the size of the example library) broaden the range of context-specific responses, and that more nuanced pattern-matching in Y corresponds to better predictions in X . That is, both variables are sensitive to this increased library size (Munch et al., 2020). If these two variables are merely correlated but not causally linked, performance will not improve beyond what might be expected from chance alone—indeed performance at small library sizes may be as good as at large. For example, if X evolves under constant Y , CCM skill should be insignificant and likely negligible when tested against a non-constant Y . This expectation is borne out by testing whether surface temperature from a transient Holocene GCM simulation with constant TSI (Tian et al., 2022) appears to bear the imprint of a TSI reconstruction (Wu et al., 2018) – it does not (see: Text S2.1, Figure S5a, and S6 in Supporting Information S1).

CCM has previously been applied to paleoclimate questions (e.g., Tsonis et al., 2015; van Nes et al., 2015), but here we take the practice of testing multiple descriptions of state (embedding configurations) and exploit it to assess this weak, uni-directional causality relationship across a wide array of timescales.

3. Experimental Design

3.1. Data

Although there is no specific rule about the minimum viable time series length for CCM, the two time series have to be long enough to supply multiple examples of various states, and resolution has to be high enough to capture the fluctuations that make those examples distinguishable. Additionally, the current implementation of CCM requires that the two time series be evenly and identically spaced. This poses a problem when trying to apply CCM in paleoclimate contexts because most records are short, with low and/or variable resolution.

To address these challenges, we leverage reconstructions of temperature and TSI that extend across the pre-industrial Holocene. We use a global mean temperature reconstruction (Erb et al., 2022, hereafter ERB22) together with a temperature reconstruction from central Greenland (GISP2; (Alley, 2000, hereafter ALLEY00)). These two contrasting data sets allow us to probe putative solar-climate connections at complementary spatial scales (global vs. local) (Figure 1a).

Further differences deserve mention. ERB22 uses data assimilation to estimate global mean surface temperature (GMST), combining a GCM-based representation of climate dynamics with proxy observations for each time slice. At each time step, model physics weave geographically dispersed observations from the Temp12k database (Kaufman et al., 2020) into a global temperature field. Although the data assimilation process introduces its own biases (Tierney et al., 2025), it provides a framework to use temperature information sourced from various types of archives (e.g., lake sediments, wood, ice cores) and observations (e.g., Mg/Ca, tree-ring width, δD) at different spatial scales, from highly local to spatially integrated. The prevalence of lacustrine and marine sediment records in ERB22 likely enhances temporal coherence but may reduce sensitivity to high-frequency climate signals better preserved in other archive types, such as ice cores or tree rings, which are scarce over the Holocene.

ALLEY00 is the central-Greenland (Summit Station; 72.6°N, 38.5°W, ~3.2 km a.s.l. (Alley, 2000)) temperature-anomaly series reconstructed from water-isotope paleothermometry ($\delta^{18}O$). The amplitude of the signal is anchored by a borehole inversion (Cuffey & Clow, 1997) and the high resolution age model pinned to annual layer counting provides chronological confidence. However, the broader climate dynamics encoded in $\delta^{18}O$ (e.g., precipitation seasonality, moisture-source shifts, circulation changes) and firm processes such as vapor diffusion (Johnsen et al., 2000) likely contribute variance not cleanly attributable to local temperature. We use the high-resolution Alley (2000) record here as the site-specific complement to the global-mean, ERB22; related Greenland temperature and $\delta^{18}O$ series assessed in parallel are summarized in Supporting Information S1.

We analyze the relationship between temperature and TSI using four dyads, reflecting the possible combinations of the two temperature estimates and two TSI reconstructions: (Vieira et al., 2011, hereafter VIEIRA11), and its update by (Wu et al., 2018, hereafter WU18). This multi-dataset approach minimizes the likelihood that our conclusions are tied to idiosyncrasies in any single reconstruction.

VIEIRA11 and WU18 (Figure 1b) use similar strategies to reconstruct TSI. Broadly, they map satellite-era TSI values to sunspot data from the instrumental period, then apply this relationship to historical sunspot observations. To extend TSI estimates into the pre-observational Holocene, they use a derived relationship between sunspot counts and production rates of cosmogenic radionuclides (e.g., ^{10}Be , ^{14}C), which vary with solar activity and are recorded in ice cores and tree rings.

WU18 updates the model used in VIEIRA11 to reflect a larger set of sunspot and radionuclide observations (^{10}Be in addition to ^{14}C) and an improved satellite calibration. These improvements are expected to reduce noise, and improve temporal resolution. Spectral analysis (Figure 1d) indicates that the canonical Gleissberg (≈ 90 years) and de Vries (≈ 200 years) cycles (Gray et al., 2010) are significant in both reconstructions. The narrower smear of significant decadal periodicities in the WU18 reconstruction may reflect the updated model. We include both because VIEIRA11 has been widely used as a solar forcing in model simulations (e.g., PMIP3 past1000 (Atwood et al., 2016); PMIP4 Holocene (Ohgaito et al., 2021)), and the WU18 reconstruction is gaining traction (PMIP4 Holocene: (Ohgaito et al., 2021)), but acknowledge that they are not fully independent.

Time series were down-scaled when necessary to a common decadal resolution (Figure 1): ALLEY00 was averaged to decadal resolution to match ERB22 and TSI. Here we consider the interval ~6100 BCE to ~1750 CE, in an effort to target a period believed to be minimally affected by residual fluctuations from the deglaciation and early influence of anthropogenic emissions.

Spectral analysis (Figures 1c and 1d) using the Multi-Taper Method with significance tested against first-order autoregressive (AR(1), i.e., red noise) null simulations at the 95% threshold ($n = 10000$) shows that both TSI reconstructions and ERB22 register significant power near the Gleissberg band (~ 80 – 100 years). ERB22 exhibits additional significant peaks aligned with TSI periodicities up to ~ 150 years (and ALLEY00 with one peak ~ 150 years). Additionally, ALLEY00 displays a distinct band of power beginning near ~ 400 years that remains barely significant through ~ 600 yr but which is not prominent in ERB22 or the TSI series.

3.2. Workflow

The CCM workflow is shown schematically in Figure 2, with further details in Text S1 in Supporting Information S1. The first step of CCM is to frame each time series as a set of embedding vectors, viz., short snippets of the parent time series that provide snapshots of the variable prior to each time step (a description of the variable's state). These snippets have two configuration parameters: E (embedding dimension), the number of data points,

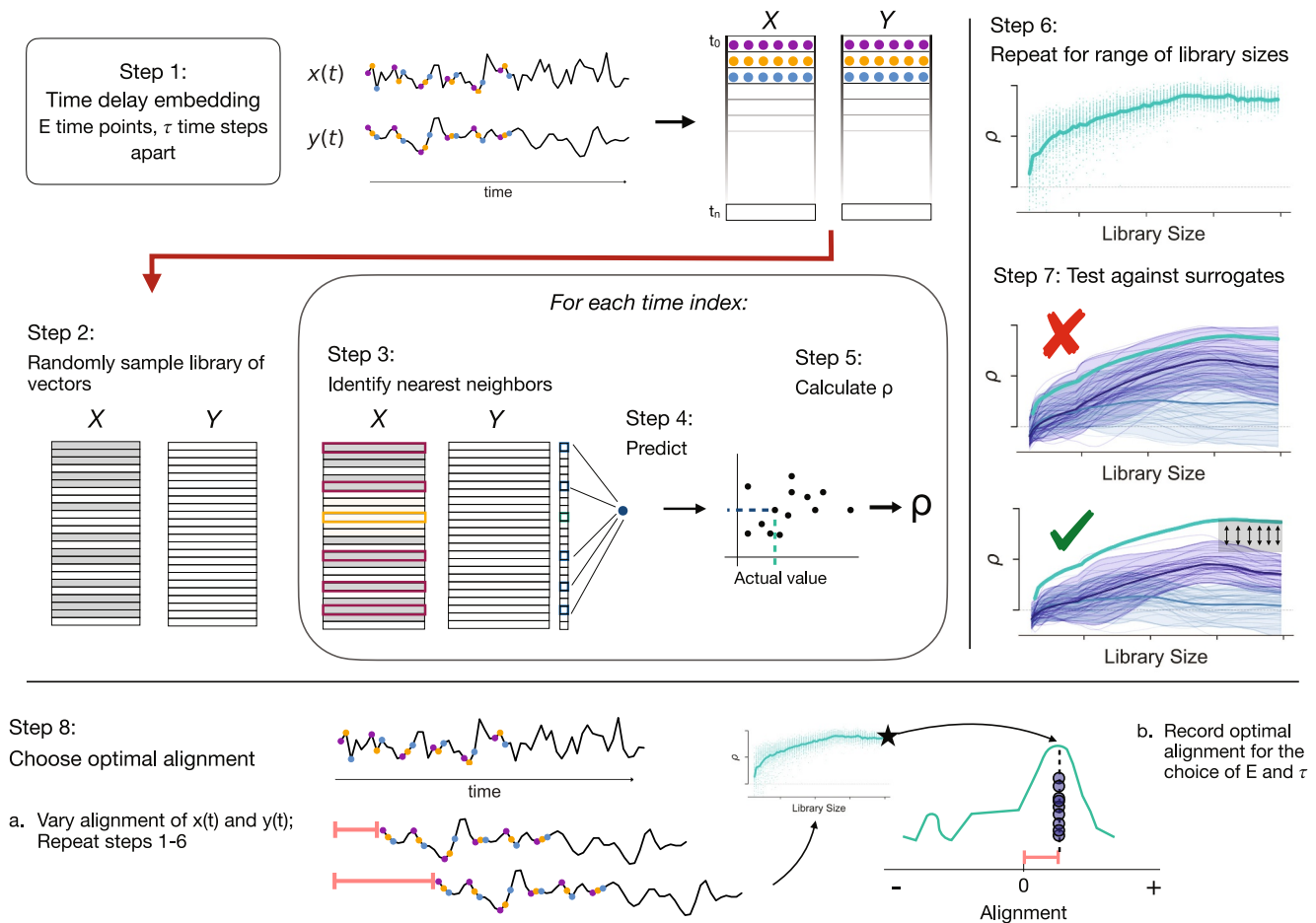


Figure 2. An overview of the Convergent Cross-Mapping workflow for assessing the causality of variable Y on X . The process involves: (1) converting time series $x(t)$ and $y(t)$ into sets of vectors, each of length E with points spaced τ time steps apart; (2) creating a “library” by randomly sampling m vectors from X (gray rows); (3) for each target point in X (a single example highlighted by the yellow box), identifying k nearest neighbors from the library (maroon boxes); (4) predicting the corresponding Y value (green box) by averaging Y values associated with the nearest neighbors (blue boxes yield blue prediction point); (5) computing the correlation (ρ) between observed and predicted values; (6) repeating this process n times across different library sizes (various values of m) to generate a convergence profile (green trajectory); and (7) evaluating significance of causality by comparing the library size- ρ trajectory of the real relationship (green trajectory) to relationships in which one variable is replaced with a surrogate (dark blue trajectory symbolizing a surrogate relationship, i.e., Y causes X surrogate, Y surrogate causes X). Additionally, (8) steps 1–6 are repeated across multiple alignments of the two time series, and the lag (ℓ) that produces the highest final ρ is recorded as the optimal value. This maximum final ρ is then used for significance testing against phase-randomized surrogates.

and τ , the number of time steps (Δt) between points. In this study, all time series are of decadal resolution ($\Delta t = 10$ years), so an embedding vector with $E = 4, \tau = 2$ spans 60 years $((E - 1) * \tau * \Delta t)$. For additional details regarding choice of E and τ , and a visual description of how embedding vectors are constructed, see, Figure S1 in Supporting Information S1.

In the context of CCM, a common approach for determining optimal embedding parameters is to use the simplex method (Sugihara & May, 1990) to identify the shared combination for which both time series best self-predict. However, initial efforts indicated that there were no clearly optimal values and that the patterns differ by variable (see: Figure S2 in Supporting Information S1). To avoid missing relationships of interest, we adopted a parameter sweep approach, considering various combinations of vector length and time spans for each dyad. Although computationally intensive, this approach provides strong coverage of the relevant timescales and is appropriate given the multi-scale nature of paleoclimate processes.

Because climate inertia and proxy archiving processes introduce lags, influence by TSI may precede its imprint in reconstructed temperature by decades or more. Following the approach of Ye et al. (2015), we compute CCM skill

(ρ) as the TSI series is shifted by ℓ decades (-40 to $+40$) where $\ell > 0$ means TSI leads temperature (forcing precedes response). For each (E, τ) we report results for the positive ℓ that yields the highest CCM skill.

To qualify as a causal relationship, a configuration must meet two conditions: (a) the predictive skill must improve as library size increases ($\rho_{\text{final}} > 0$ and $\Delta\rho = \rho_{\text{final}} - \rho_{\text{initial}} > 0$; here, “initial” refers to the smallest library size; “final” to the largest library size) and (b) the skill must converge (or show a clear tendency toward convergence) to a stable ρ value. Convergence indicates that the relationship has been well sampled, that addition of further examples will not substantively improve prediction skill.

To be considered significant, a causal relationship must demonstrate predictive skill beyond what would be expected from random chance (Deyle, May, et al., 2016; van Nes et al., 2015). To assess significance, we compared each TSI–temperature relationship (“real relationship”) to 200 distinct examples of the corresponding case in which either the TSI or temperature reconstruction was replaced with a phase-randomized surrogate (“surrogate relationship”) (Prichard & Theiler, 1994). This method of generating surrogates preserves the spectral content of the records while moving their features around in time. For a given (E, τ) configuration, a relationship is considered significant if the final ρ of the real relationship exceeds the 95th percentile of final- ρ values for each category of surrogate relationship. To count as a significant climate driver, TSI series should show both causal and significant associations with temperature in a clear majority of (E, τ) configurations.

4. Results

We evaluated the relationship “TSI influences temperature” by examining 49 (E, τ) configurations across each TSI–temperature pairing. Each relationship configuration was evaluated for over 200 samplings for each of a range of library sizes, for both the real relationship and again in the context of 400 surrogate relationships. To efficiently and objectively summarize these high-dimensional results, we present summary statistics organized in four grids (Figure 3). A detailed discussion of intermediate steps is available in Text S1 in Supporting Information S1.

Each of the four grids in Figure 3 represents a different combination of temperature target (ALLEY00 or ERB22) and TSI reconstruction (WU18 or VIEIRA11). The color of each grid box indicates improvement in CCM skill ($\Delta\rho$) when evaluating the proposition “TSI had a discernible causal influence on the target temperature series” using an E (column value)– τ (row value) configuration at its selected lag, ℓ . Broadly, Figure 3 shows that all configurations of each dyad register improvement ($\Delta\rho > 0$) with increasing library size—a necessary condition for causality. Importantly, the majority of configurations are significant when tested against surrogates, providing evidence that TSI leaves a discernible imprint on temperature. Both $\Delta\rho$, an empirical measure of the strength of evidence, and significance show diagonal structure indicating results are organized according to increasing snippet time span, $(E - 1) * \tau * \Delta t$.

$\Delta\rho$ generally increases with longer snippet time spans in ERB22–WU18 and to a lesser extent in ERB22–VIEIRA11, producing diagonal color gradients in the results grids. Almost all ERB22 configurations remain significant under surrogate testing, with a single exception coming from ERB22–VIEIRA11 ($E = 10, \tau = 7$). ERB22–WU18 $\Delta\rho$ surpasses ERB22–VIEIRA11 $\Delta\rho$ in most configurations, though differences are modest. In contrast, the timescale structure is stronger in the ALLEY00 dyads. In each case, improvement is high for spans below and above a narrow valley of insignificance centered near ~ 300 years. ERB22 results tend to be associated with ℓ values close to 0 and record higher ρ at the lowest library sizes resulting in smaller $\Delta\rho$, whereas ALLEY00 results are reported for ℓ values between 10 and 20, start lower and gain more (see: Figure S7 in Supporting Information S1).

5. Discussion

Application of the CCM framework to four TSI–temperature reconstruction pairings suggests that TSI exerted a discernible influence on temperature over the Holocene on multi-decadal to millennial scales, according to 80% of the parameter combinations examined. This supports the proposition that solar variability contributed to preindustrial Holocene climate variability. Although this relationship has long been hypothesized from regional studies (e.g., Denton & Karlén, 1973; Renssen et al., 2006; Turney et al., 2005) or those focused on particular periods (e.g., the Holocene Climate Optimum, (Debret et al., 2007) or the Little Ice Age, (Lean & Rind, 1999)), it had not been empirically substantiated through comprehensive analysis.

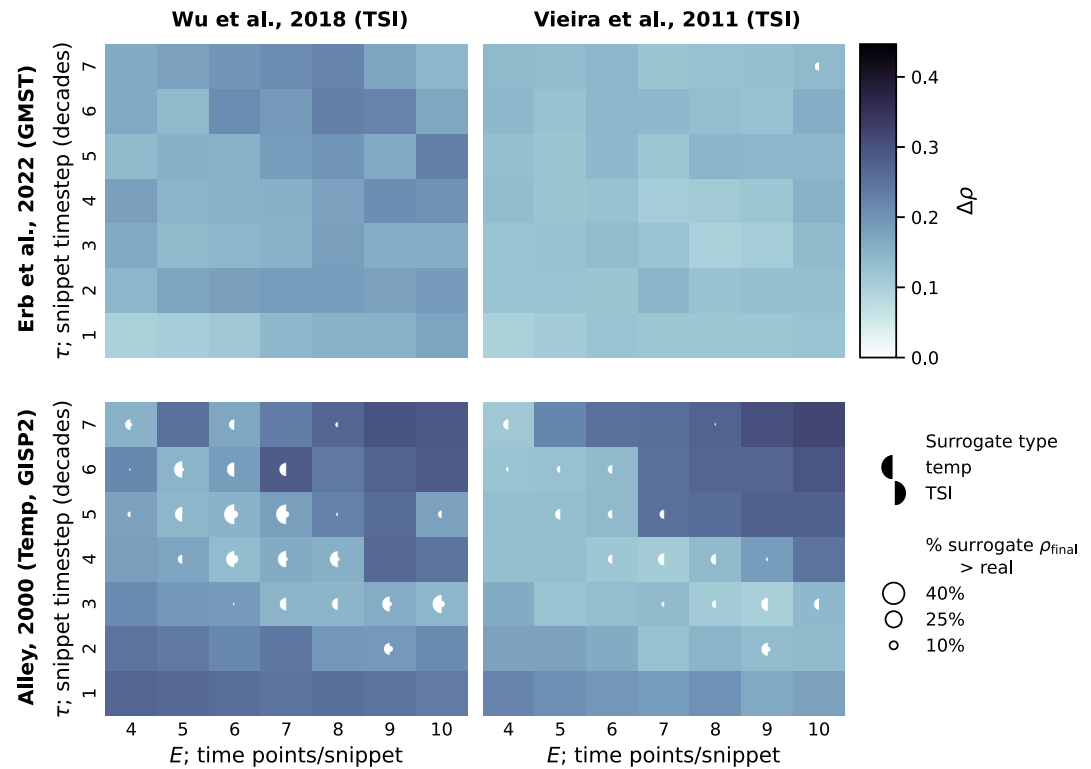


Figure 3. Summary of metrics assessing significant causality for the relationship “TSI influences temperature” using Convergent Cross-Mapping across all combinations of embedding dimension E (columns) and resolution τ (rows) for each TSI-temperature dyad. Subpanels are organized by temperature target (ERB22 top, ALLEY00 bottom) and Total Solar Irradiance source (WU18 left, VIEIRA11 right). Each grid box summarizes one (E, τ) configuration at the lag ℓ that maximizes skill for that configuration: The results of each configuration are summarized by the contents of the individual grid boxes: (1) color corresponds to improvement in ρ for the real relationship, $\Delta\rho (\rho_{\text{final}} - \rho_{\text{initial}})$, (2) size of half-moons indicate surrogate failure rates (percent of surrogate relationships with final $\rho \geq \text{real final } \rho$); left half-moons for “TSI influences temperature (surrogate)” and the right for “TSI (surrogate) influences temperature.” Significance is evaluated on final ρ against phase-randomized surrogates as defined in Methods.

The TSI–temperature relationship is broadly detectable, but its heterogeneity in data sets and configurations (E, τ) is consistent with a subtle solar imprint modulated by the characteristics of the scale and the record, rather than a dominant influence. ERB22 (multi-archive, global) tells a story that is consistent with ALLEY00 (single, high-latitude site) at a high level, but differs in the nuance. Global averaging and an emphasis on long, overlapping records in ERB22 yield more coherent low-frequency structure, raising overall predictability (higher and more consistent final ρ) and damping incremental gains (smaller $\Delta\rho$, especially at short timescales). A single site can preserve a tighter response (expressed as a greater $\Delta\rho$) because less spatial averaging reduces cancellation and local traits may improve sensitivity to the relationship. For example, in the case of ALLEY00, cryosphere feedbacks amplify temperature variability, and the high-altitude interior location of GISP2 is hypothesized to shield it from maritime air masses and the thermal buffering influence of ocean heat capacity, which tends to damp temperature variability at lower-altitude and coastal sites (Hébert et al., 2022).

The narrow loss of significance near ~ 300 years in the ALLEY00 grids occurs when the snippet time span overlaps the multi-century band where ALLEY00 shows modest but significant periodic structure (Figure 1). In a nonlinear system, the more periodic a response, the more regular and the less distinctive its system states, and the harder it is for CCM to distinguish real from surrogate relationships. In this situation, phase-randomized surrogates become more difficult to distinguish from the real relationship because periodicity reduces the distinctiveness of trajectories in that frequency band (c.f. Deyle, Maher, et al., 2016), not because causal influence is absent. Thus, this increased periodicity can lead to both predictability and insignificance, due to this limitation of surrogate-based significance testing in near-periodic regimes that is not indicative of an absence of nonlinear coupling.

It is critical to put the conclusion that “TSI had a discernible influence on Holocene climate” into context. The evidence for a causal link between TSI and climate over the Holocene does not challenge the dominant role of anthropogenic CO₂ during the Industrial Era, during which the radiative forcing from human activities far exceeds that of solar variability. Indeed, relative to a pre-industrial baseline, the Intergovernmental Panel on Climate Change estimates that net anthropogenic radiative forcing increased by approximately 2.72 W/m² (with greenhouse gases being the largest contributor), while solar forcing has remained nearly constant (estimated change of −0.08 to 0.06 W/m²) (Intergovernmental Panel on Climate Change (IPCC), 2023).

The sensitivity of our result, produced using a non-mechanistic approach, aligns with the broader difficulty faced by mechanistic models, both low-order (Emile-Geay et al., 2007) and comprehensive (Ziegler et al., 2024), in generating substantial climate shifts from solar forcing alone. Climate simulations of the past millennium indicate that, in the absence of anthropogenic factors, even an extreme increase in solar irradiance combined with volcanic effects failed to produce late-20th-century warming (Otto-Bliesner et al., 2015). Because CCM evaluates directional predictability rather than the amplitude of the forced response, it can register an empirical TSI–temperature link even when a forced temperature signal is too small to emerge in models. Our findings provide an observation-based line of evidence for this relationship, but do not contradict the muted role of solar forcing seen in physics-based models. Rather, this confirms that TSI is one of several ingredients of climate variability, and one that could potentially be leveraged to improve predictions on multi-decadal timescales.

Approaching this study as a parameter sweep and using multiple data set pairings was critical to minimizing the influence of confounding factors in the TSI–temperature relationship. In the physical world, independent, stochastic forcings, such as explosive volcanism, are known to influence temperature, with climatic responses typically on the order of a few years. However, in decadal-resolution proxy records with inherent age-model uncertainty, these transients can be aliased, and present as broader climate trends, altering the perceived TSI–temperature relationship. Similarly, internal climate modes operating at timescales akin to those that appear in TSI (e.g., ENSO, NAO, and AMO) can also complicate interpretation by affecting temperature in patterns that mimic TSI periodicities, while actually being only distantly connected to TSI, if at all. Though surrogate testing cannot identify specific sources of spurious signal, it does provide a check as to whether such processes could fool the framework, and if so, on what timescales. Slowly evolving background conditions such as insolation can be similarly confounding, introducing subtle macro-structure (e.g., slight trend in ALLEY00 temperature in Figure 1a); to this end, we repeated our experiments on linearly detrended and 1kyr high-pass filtered versions of all dyads and though the rate of significance was lower, a similar pattern of results persisted (see: Text S2.2, Figures S5, and S6 in Supporting Information S1).

Apart from climatic factors, methodological choices in the generation of these records may further complicate the picture, altering the pattern of uncertainty and, in the case of TSI, reducing independence of the reconstructed time series. ERB22, WU18, and VIEIRA11 all combine multiple inputs through a model or inversion. ERB22 uses data assimilation to knit together a diverse pool of proxy records (Kaufman et al., 2020). VIEIRA11's solar-activity precursor is a single ¹⁴C-based open-flux series, while WU18's is a Bayesian composite of ¹⁰Be and ¹⁴C records, both converted to TSI using a shared irradiance model. In contrast, ALLEY00 is a single-site temperature estimate from a central Greenland ice core made with water-isotope paleothermometry and minimally constrained by borehole inversions. Unlike the others, however, it is subject to, and preserves site-specific influences of ice-core processes (e.g., snowfall seasonality, post-depositional effects, shifts in moisture source), which introduce features not representative of GMST and complicate direct comparison to the global signal.

Continued work on Greenland ice cores has produced a broader set of high-quality temperature and δ¹⁸O reconstructions. We repeated our analysis for several records distributed between GISP2 and NGRIP (Döring & Leuenberger, 2022; Martin et al., 2024; Seierstad et al., 2014) (see: data sets: Text S2.1, Figure S5, extended results: Figure S6 in Supporting Information S1). Despite differences in measured quantity and reconstruction methodology, the overall picture is similar to that from ERB22-GMST and ALLEY00-δ¹⁸O-derived Greenland temperature. Dyads that pass the CCM criteria show a pattern: non-δ¹⁸O air-temperature reconstructions (GISP2 from Döring and Leuenberger (2022), and ERB22-GMST) yield optimal lags close to zero, whereas δ¹⁸O series (GISP2 and NGRIP from Seierstad et al. (2014)) and δ¹⁸O-derived temperature (ALLEY00) tend to favor positive lags of about 100–200 years (10–20 time steps). CCM identifies directional predictability between variables, not the physical mechanisms by which that influence operates, and we do not interpret these centennial offsets as a

deterministic response time to TSI. Rather, the contrast between near-zero lags in temperature reconstructions and longer lags in $\delta^{18}\text{O}$ records is consistent with $\delta^{18}\text{O}$ integrating local atmospheric temperature with slower processes linked to ocean circulation, sea surface temperature, sea ice extent, and precipitation sources, as previously suggested (e.g., Seierstad et al., 2014). Detecting that TSI leaves a discernible imprint on temperature is not equivalent to identifying how it does so; these are complementary but separate notions.

Finally, we note that our nonlinear framework assesses causality, but cannot quantify the fraction of temperature variance explained by TSI (an inherently linear notion). Sensitivity of the strength of evidence and significance to data set choice and configuration suggest that TSI was not the primary driver of preindustrial Holocene climate variability. If TSI were a dominant or direct driver of climate variability, a simple linear relationship would be expected and recoverable by simple correlation. However, such consistent significant correlation is not present in these data sets (see: Text S3, Figure S8 in Supporting Information S1). The pervasive insignificance among simple correlation results and superior performance by CCM (Figure S10 in Supporting Information S1) underscore the complexity of isolating a solar influence from available records amid other climate signals, and the need to consider this relationship with a nonlinear lens.

From the viewpoint of this study, TSI variability appears likely to remain a minor modulator of climate over the next century relative to anthropogenic forcing. Despite the subtlety of the influence, our results indicate that this relationship is recoverable when it is present, an outcome that could be leveraged by climate models forced with variable TSI as a benchmark of model physics.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

Data sets used here are publicly available as cited. CCM outputs are archived on Figshare (v1.1) (Landers, Emile-Geay, James, et al., 2026). The analysis library Cedarkit (v2026.05.02b) is available on Zenodo (Landers, 2026) and Github (https://github.com/jordanplanders/cedar_util), and code to reproduce figures (v0.0.1b) from this study will be available via the PaleoBook Gallery (<http://linked.earth/PaleoBooks>) upon publication (Landers, Emile-Geay, Khider, & James, 2026).

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